



Foundations of Programming

Assignment No.: 18

PROBLEM STATEMENT: Write a C program that accepts a string from the user and performs the following string operations: i) Calculate length of string ii) String reversal iii) Equality check of two strings iv) Check palindrome v) Check substring.

OBJECTIVE:

- To understand basic string operations in C.
- To learn how to manipulate strings using loops and functions.
- To perform comparison, reversal, and substring checking.
- To check whether a string is a palindrome.

THEORY:

In C programming, a string is a sequence of characters terminated by a null character ('\0'). Strings are commonly stored in character arrays. Various operations can be performed on strings such as calculating their length, reversing the characters, comparing two strings, checking whether a string is a palindrome, and finding a substring within another string. These operations can be implemented using loops and conditional statements or by using standard library functions from the string.h header file.

PLATFORM: Linux

CODE:

```
#include<stdio.h>
```

```
#include<string.h>
```

```
int main()
```

```
{
```

```
    char str1[100], str2[100];
```

```
printf("Enter first string: ");
scanf("%s",str1);

/* 1. Length */
printf("Length = %lu\n", strlen(str1));

/* 2. Reverse */
char rev[100];
strcpy(rev,str1);
strrev(rev);
printf("Reverse = %s\n", rev);

/* 3. Equality check */
printf("Enter second string: ");
scanf("%s",str2);

if(strcmp(str1,str2)==0)
    printf("Strings are Equal\n");
else
    printf("Strings are Not Equal\n");

/* 4. Palindrome check */
if(strcmp(str1,rev)==0)
```

```

        printf("Palindrome\n");
    else
        printf("Not Palindrome\n");

    /* 5. Substring check */
    if(strstr(str1,str2)!=NULL)
        printf("Substring Found\n");
    else
        printf("Substring Not Found\n");
}

```

INPUT:

```

Enter first string: madam
Enter second string: ada

```

OUTPUT:

```

Length = 5
Reverse = madam
Strings are Not Equal
Palindrome
Substring Found

```

SAMPLE INPUT:

```

Enter a string: madam
Enter another string for equality check: mad

```

SAMPLE OUTPUT:

```

Length of string = 5
Reversed string = madam

```

The string is a Palindrome
Strings are not equal
Substring found in main string

CONCLUSION: Thus, the C program successfully performs different string operations such as length calculation, reversal, equality check, palindrome check, and substring detection.

FAQs:

1. What is a string in C programming?
2. Which header file is required for string functions?
3. What function is used to find the length of a string?
4. What is a palindrome string?
5. Which function is used to check a substring in C?